जम्मू केंद्रीय विश्वविद्यालय

Central University of Jammu

रह्या-सुचानी (बगला), जिला सांबा -181143, (जम्मू और कश्मीर)

Rahya-Suchani (Bagla), District: Samba – 181143, Jammu (J&K)

24th July 2025

Roll No. | | | | | | | | | | | | |

SEMESTER MAJOR EXAMINATIONS – MAY 2025

Programme	:	B.Tech. CSE	Semester	:	VI
Course Name	:	Big Data Analytics	Course Code	:	BECSE3C007
Duration	:	**3 Hours**	Credits	:	**03**
Maximum Marks	:	75	Minimum Marks	:	

After pursuing this course, the student will be able to:	
CO 01	Recall the fundamental concepts of Big Data and explain its role in addressing business challenges.
CO 02	Describe the components of Hadoop and evaluate the functionality of tools in the Hadoop ecosystem.
CO 03	Demonstrate the ability to access, process, and manage data using distributed file systems like HDFS.
CO 04	Design and develop efficient Big Data solutions by leveraging Hadoop and its associated tools.
CO 05	Access data analysis challenges and implement machine learning models.

Part A **Very Short Answer Type Logical Questions of 03 marks related to different COs** **(All the units to be covered from the entire syllabus)**		**Total Marks 3X5=15**	
S.No.	**Statement**	**Levels**	**CO**
1.	Define Big Data and explain its characteristics.	L1	CO-01
2.	List the key components of Hadoop and briefly describe their functions.	L1	CO-02
3.	Explain the concept of Data Cleaning and its importance in Data Analytics.	L2	CO-03
4.	Describe the role of NoSQL databases in Big Data management.	L2	CO-04
5.	Mention the AI Models in Data Cleaning, and discuss about the Error Detection & Correction steps involved for Big Data.	L2	CO-05
PART B **Short Answer Type Analytical Questions of 6 marks related to different COs** **(All the units to be covered from the entire syllabus)**		**Total Marks 6X5=30**	
6.	State the Differences between SQL and NoSQL Based on Architecture and Operations.	L4	CO-01
7.	Compare and contrast the MapReduce programming model with traditional data processing techniques.	L4	CO-02
8.	How does Exploratory Data Analysis (EDA) help in understanding datasets? Demonstrate with a Python example.	L3	CO-03
9.	Describe Collaborative Filtering and explain its role in recommendation systems. Also draw the diagram for the same.	L4	CO-05
10.	Write a Python program to clean and preprocess a dataset for analysis. Explain the steps involved.	L3	CO-05

Part C Long Answer Type Descriptive Questions of 15 marks related to different Cos (Only Two questions to be attempted)		Total Marks 15X2=30	
11.	Assess the scalability challenges in Big Data analytics and propose strategies to optimize performance in a distributed computing environment. Analyze the impact of Big Data in real-world business scenarios and decision-making.	L5	CO-01,03
12.	Explain how Brandwatch collects social media data. Identify and analyze three major challenges that Brandwatch and similar tools face while processing social media data. Provide relevant examples for each challenge. Evaluate the impact of data privacy regulations such as GDPR and CCPA on the data collection and usage policies of social media analytics tools. How do these tools ensure compliance while still providing valuable insights?	L6	CO-05
13.	Explain the key differences between Supervised and Unsupervised Learning. Support your explanation with appropriate real-world examples in Big Data Analytics. Compare and contrast the interpretability of Supervised vs. Unsupervised Learning models when dealing with large-scale retail datasets.	L6	CO-04,05